Headphones, No Headphones

An Informal Observational Study of Solo Pedestrian Audio Device Usage

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Abstract

We aim to study how many solo walkers are wearing headphones versus not wearing headphones. Data were collected across ten observation sessions in Madison, WI between February and November 2024. Across all sessions, a substantial majority of solo walkers were observed wearing headphones. No inferential statistics were calculated; the data are presented for descriptive and exploratory purposes only.

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1. Terms

Headphones: For the purposes of this study, any personal audio output device (eg. earbuds, over-ear headphones, on-ear headphones, etc.) will be referred to as "headphones."

2. Inclusion and Exclusion Criteria

- If somebody is running, they are not counted.
- If somebody is on a bike, scooter, in a car, etc., they are not counted.
- If it cannot be determined whether somebody is wearing headphones, they are not counted.
- If a person is in a group, meaning they are conversing with or otherwise clearly out with another person or persons, they are not counted.
- If a person's clothing, hair, or accessories (including ear muffs, under which earbuds could plausibly be worn) prevented clear and conclusive determination of headphone status, they were not counted.
- If somebody has headphones around their neck but not on their ears, they count as no headphones.
- If somebody has a single earbud in, they count as headphones.

Data were only collected from people walking to or from a place, not from individuals stationary at a location (ie. no counting people working out at a gym, sitting in class, sitting at work, etc.)

3. Limitations of Data Collection

Many individuals wear a single earbud. If that earbud was visible, they were counted toward

“headphones,” but it is plausible that some individuals labeled as “no headphones” had a non-visible earbud in.

It is easier to conclusively identify headphone usage than its absence. For example, if a person with long hair has full-size headphones on, that is easy to identify. But if that same person did not have headphones on, they might not be counted at all, as it could be impossible to determine whether they were wearing earbuds.

A single person collected all data on various walks. The data are inherently imperfect, as they reflect a subjective human observer recording information from memory. Data were recorded in the researcher’s mind during collection and written down only at the conclusion of each walk. Human memory is known to be error-prone [1][2].

4. Data

Table 1 presents raw counts from all ten observation sessions. Across all sessions, 444 walkers were observed wearing headphones and 209 without, yielding a headphone prevalence of 68.0%.

Table 1. Observation session data.

Location	Observation Window	Headphones	No Headphones
Madison, WI	February 20, 2024 9:15am – 9:45am	50	22
Madison, WI	February 20, 2024 1:30pm – 1:40pm	14	6
Madison, WI	February 21, 2024 3:50pm – 4:00pm	33	17
Madison, WI	February 27, 2024 2:25pm – 2:30pm	14	3
Madison, WI	April 4, 2024 2:10pm – 2:25pm	83	21
Madison, WI	April 17, 2024 9:45am – 9:55am	33	15
Madison, WI	September 10, 2024 3:40pm – 3:50pm	33	27
Madison, WI	September 10, 2024 5:20pm – 5:30pm	48	32
Madison, WI	September 30, 2024 1:30pm – 2:00pm	59	30
Madison, WI	November 13, 2024 2:00pm – 2:30pm	77	36
Total	**All sessions**	**444**	**209**

5. Visualizations

The following figures present the data graphically. No statistical significance tests were performed; these visualizations are intended for descriptive and exploratory purposes only. Session counts and observation durations vary across periods, and comparisons should be interpreted accordingly.

5.1 Overall Totals

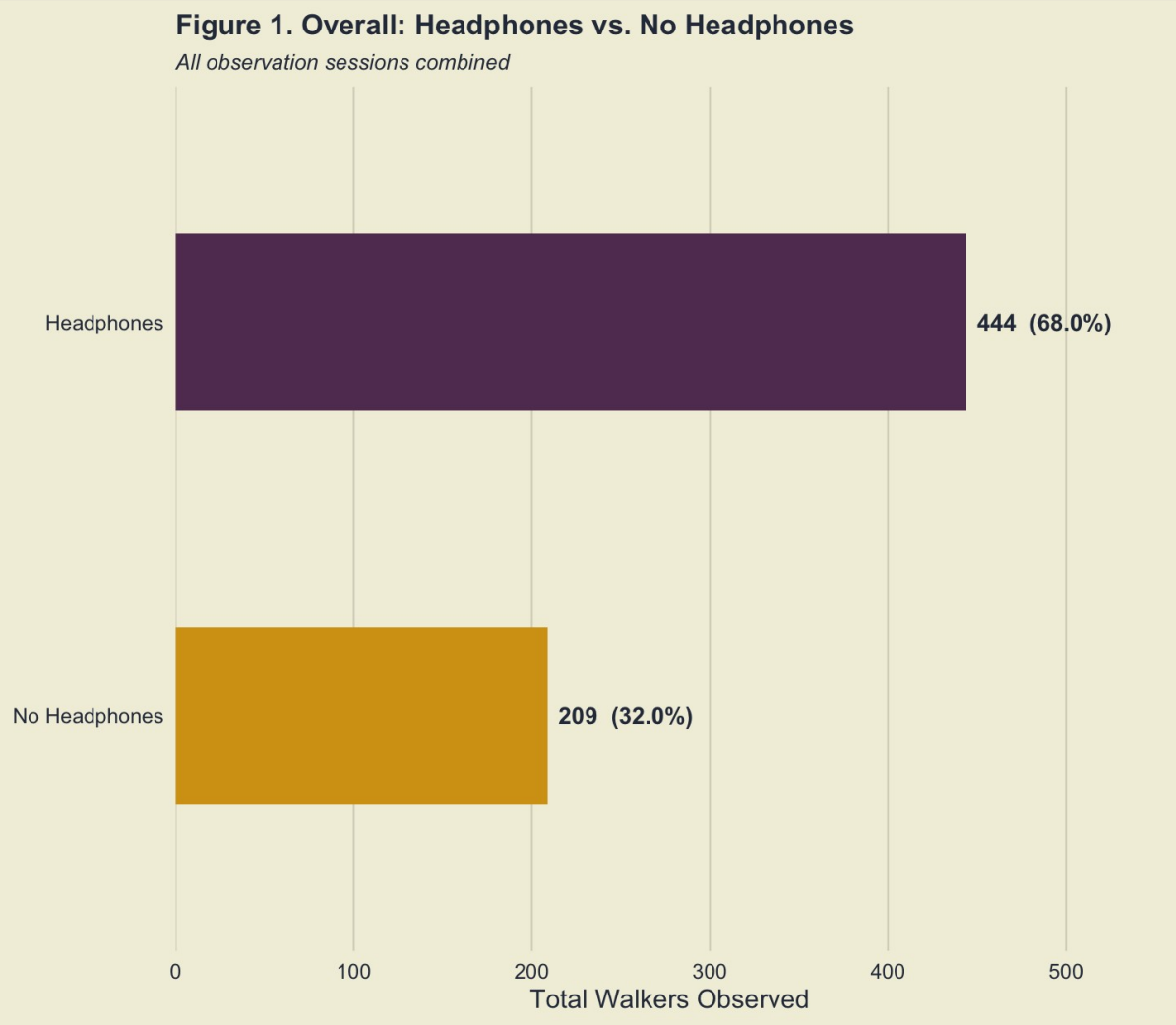

Figure 1. Total walkers observed across all sessions, split by headphone status.

5.2 Per-Session Breakdown

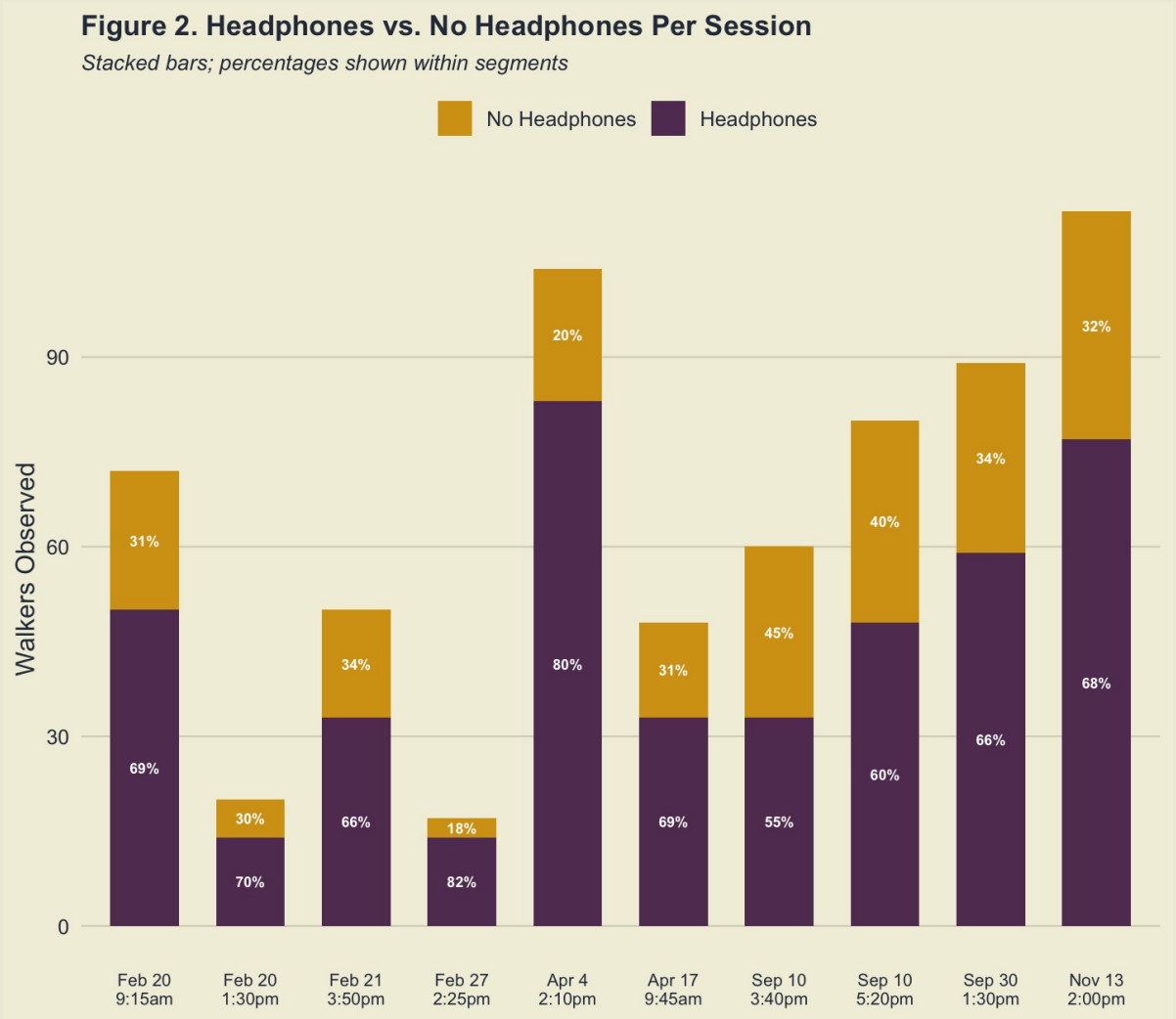

Figure 2. Stacked bar chart of headphone vs. no headphone counts per observation session. Percentages are

shown within segments.

5.3 Day of the Week Comparison

For exploratory interest, sessions were aggregated by day of the week. Session counts per day differ substantially (range: 1–5 sessions), so apparent between-day differences carry very limited interpretive weight. No statistical tests were run.

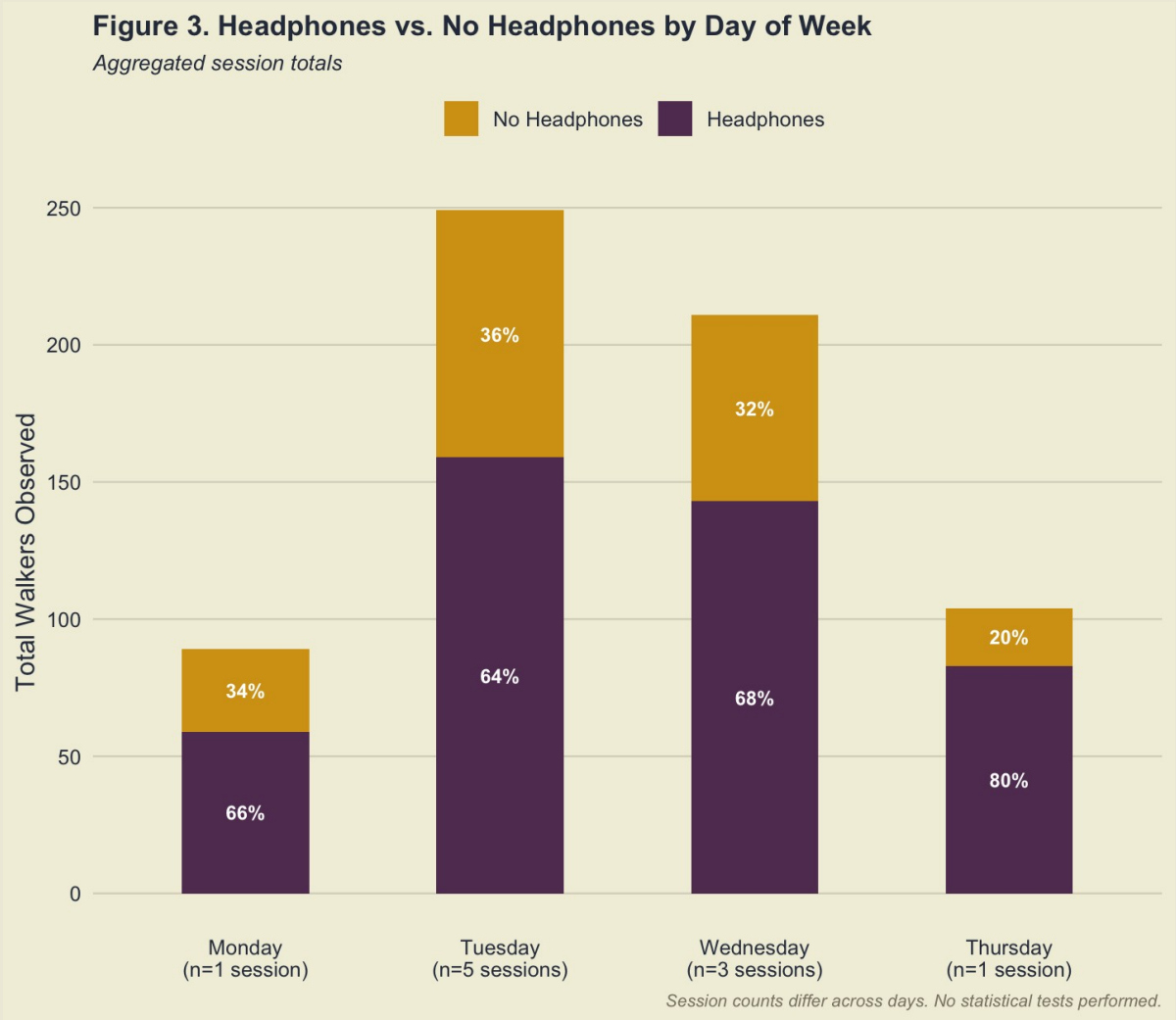

Figure 3. Aggregated headphone and no-headphone totals by day of week. Statistical significance was not calculated.

6. Bibliography

[1] D. L. Schacter, S. A. Guerin, and P. L. St. Jacques, "Memory distortion: an adaptive perspective," *Trends in Cognitive Sciences*, vol. 15, no. 10, pp. 467–474, Oct. 2011, doi: https://doi.org/10.1016/j.tics.2011.08.004. Available:

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3183109/

[2] M. F. Panichello, B. DePasquale, J. W. Pillow, and T. J. Buschman, “Error-correcting dynamics in visual working memory,” *Nature Communications*, vol. 10, no. 1, Jul. 2019, doi: https://doi.org/10.1038/s41467-019-11298-3